# Janusan Jeyananthan

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## Education

**University College London** *Sep 2022 – July 2026*
*MSci Biochemistry*

- **Relevant Modules:** Computational and Systems Biology, Linear Algebra for Data Science, Practical and Computational Biomolecular Structure and Function, Regulatory Genomics and Evolution

## Research Experience

**MSci Research Thesis** *Oct 2025 – April 2026*
**Density-based clustering and contrastive fine-tuning of protein language model embeddings for scalable functional family generation**
*University College London* Supervisor: Professor Christine Orengo

- Developed and evaluated a computational pipeline applying density-based clustering (HDBSCAN) to protein language model embeddings for large-scale protein functional family generation, achieving comparable or higher functional purity than existing approaches while demonstrating a >1000× reduction in runtime
- Trained and evaluated supervised contrastive learning models on a 3M+ sequence dataset to fine-tune pre-trained embeddings for improved functional clustering
- Thesis work is being further developed through ongoing research with the goal of publication

**Undergraduate Research Project (3rd Year)** *Sept 2024 – June 2025*
*University College London*

- Carried out 16S rRNA amplicon sequencing analysis to profile sink microbiomes across hospital, university, and domestic environments
- Assessed alpha/beta diversity and taxonomic composition to evaluate environmental impacts on microbial communities
- Produced written report detailing experimental data preparation, data analysis and findings

**Undergraduate Literature Dissertation (3rd Year)** *Sept 2024 – June 2025*
*University College London* Supervisor: Professor Christine Orengo

- **Title:** Computational methods for predicting and analysing protein-protein interaction (PPI) networks
- Authored a comprehensive review of methodological advancements, limitations, and future directions for PPI prediction and network analysis with a focus on recent deep learning based methods
- Presented findings in an assessed oral presentation to staff and students

## Publications

**ContrasTED: Resolving protein dark matter with supervised contrastive learning** 2026
D.Miller, N.Bordin, **J.Jeyananthan**, V.Waman, M.Heinzinger, C.Orengo. *Under review*

## Technical Skills

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| **Programming & Deep Learning** | Python, PyTorch, experiment tracking (Weights & Biases) |
| **Bioinformatics** | Sequence search and clustering (MMseqs2, BLAST), protein structure prediction and analysis (AlphaFold 2, Foldseek, PyMol), 16S rRNA amplicon sequencing workflows (VSEARCH) |
| **Data Analysis & Statistics** | Pandas, data visualisation (Matplotlib) |
| **Computing** | HPC, Linux, Git |

## Awards

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| iGEM Team Gold Medal — City of London School | 2021 |
| Academic Scholarship — City of London School | 2021 |
| Gold Award — UKMT Intermediate Maths Challenge | 2019 |

## Additional Skills & Interests

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| **Languages** | English, Tamil (native proficiency); French, Spanish (working proficiency) |
| **Activities** | UCL Boxing Competitive Squad (2022–2024); Hydrogen Dance Company (2025–present) |